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Demographic Trends and Transition in Odisha: Emerging Patterns and Implications

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Abstract

The Indian state of Odisha has seen remarkable demographic changes towards the end of the 20th century and the last one and a half decade of 21st century. During the same period, the state has also experienced socio economic transformations of varying degree. Demographic situation in Odisha was characterised by high birth rate and death rate in 1971. As per the National Family Health Survey (NFHS 4, 2015-16), Odisha has achieved replacement level of fertility. On the contrary, the mortality in the early years of childhood remains substantially high in spite of a declining trend. The demographic history of Odisha throws a number of surprises, one being the situation of high level of childhood mortality and low level of fertility. A perfect portrayal of demographic transition requires an investigation of the impact of demographic changes on socio-economic transformations and vice-versa. This paper examines the trends and regional heterogeneities in demographic scenario in the state by using the data from the Census of India, the Sample Registration System (SRS) and the National Family Health Surveys (NFHS).

Keywords: demographic transition, fertility, mortality, age structure, population growth

Introduction

Demographic history demonstrates that countries, particularly of Western Europe and North America moved from a regime of high mortality and high fertility to a regime of low mortality and low fertility. This shift has been conceptualised as 'demographic transition' which is defined as the process of secular shift in fertility and mortality from high and fluctuating levels towards a presumed low and relatively stable levels. It was presumed that rest of the countries would follow a similar kind of

shift in demographic situation. India, the second populous country in the world with a population size of 1.3 billion has experienced a demographic transition too. With Crude Death Rate (CDR) and Crude Birth Rate (CBR) above 40 during the early years of 20th century, India has seen a dramatic decline in these two vital events. According to latest figures of the Sample Registration System (SRS), CBR and CDR in India were 20.4 and 6.4 respectively in 2016 (Registrar General of India, 2016). Overall demo-

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graphic trends reveal that India is transitioning from late expanding to low stationary stage of demographic transition (Bhat and Rajan, 1997; Dyson, 1997; Guilmoto and Rajan, 2013; Visaria, 2011; Kulkarni, 2011). Given the demographic diversity of the country, it is not surprising to find different states of the country at different stage of demographic transition (Bhat and Rajan, 1990; Krishnana, 1976). It is widely known that the crux of India's population problem lies in the north-central India states such as Bihar (including Jharkhand), Madhya Pradesh (including Chhattisgarh), Rajasthan and Uttar Pradesh (including Uttarakhand). Prof. Ashish Bose had coined the acronym 'BIMARU' (in Hindi bimaru means sick) formed from the first letters of the names of these states, lagging behind in the demographic transition with poor demographic and health indicators. Because of higher Infant Mortality Rate in Odisha, 'O' (for Odisha) was added to 'BIMARU' later on culminating in the acronym 'BIMARUO'. Owing to the derogatory meaning of the term 'BIMARU', officially this group of states has been known as Empowered Action Group (EAG) States. Currently, there are eight (Bihar, Chhattisgarh, Jharkhand, Madhya Pradesh, Odisha, Rajasthan, Uttaranchal and Uttar Pradesh) EAG states in India.

What has happened to the eastern Indian state of 'Odisha' with a population size of 41.9 million? Odisha has attracted the attention of population scientists for a long time now, the reason being it has achieved a lower level of fertility in the absence of similar achievement in mortality and the general socioeconomic conditions. The impressive performance of Odisha state in the fertility front has received worldwide attention and admiration. The interest was obviously generated from the fact that in terms of per capita income and industrial output Odisha ranks among the poorest states in India. In 2011-

12, the percentage of people Below Poverty Line (BPL) was 32.59 % (Government of India, 2013). Many demographers call this situation as demographic dilemma. This paper attempts to chat out the recent demographic history of Odisha by a careful scrutiny of evidence presented by various data sources and identify the key the demographic concerns highlighting their future socio-economic implications.

Database and methodology

This paper is based on the data from the Census of India, the Sample Registration System (SRS) and the National Family Health Survey (NFHS). Two rounds of National Family Health Surveys were conducted in 1990's; NFHS I was conducted in 1992-93 and NFHS 2 was conducted in 1998-99. The third and fourth rounds of NFHS were conducted in 2005-06 and 2015-16 respectively. This source of data has been utilised to look into the change in fertility and mortality indicators in the recent past. However, the Sample Registration of India provides fertility and mortality indicators from the early 1970s. For the period 1950-60, estimates of birth rate and death rate from report on vital statistics and fertility survey, published as part of census of India, 1961 have been used. Analysis of population growth and age structure is based on census of India data. Different demographic indicators such as Crude Birth Rate (CBR), Total Fertility Rate (TFR), Total Wanted Fertility Rate (TWFR), Crude Death Rate (CDR), Infant Mortality Rate (IMR), Under Five Mortality Rate (UFMR), Maternal Mortality Rate and Life Expectancy are taken into consideration to portray the demographic changes. MsExcel has been used to analyse data and preparing diagrams while maps have been prepared through ArcGIS software. The definition and measurement of indicators are as follows:

The Crude birth rate (CBR) is the number of live

births per 1000 mid-year population in a given geographical area during a given year.

The total fertility rate (TFR) The number of children who would be born per woman if she were to pass through the childbearing years bearing children according to a current schedule of age-specific fertility rates.

The crude death rate (CDR) is the total number of deaths per 1000 mid-year population in a given geographical area during a given year.

The infant mortality rate (IMR) is the number of deaths of infants under one year old per 1000 live births in a given year.

The maternal mortality rate (MMR) is the annual number of female deaths per one lakh live births from any cause related to or aggravated by pregnancy or its management.

Life expectancy at birth is the average number of years that a newly born baby is expected to survive if current mortality rates continue to apply.

The dependency ratio is a measure showing the ratio of the number of dependents aged

zero to 14 and over the age of 65 to the total population aged 15 to 59.

Locating Odisha in the demographic maps of India

Table 1 provides an overview of Odisha's performance in terms of socio-demographic and health indicators in national context. As per the 2011 census, the population of Odisha is 41.9 million. It experienced a decadal growth rate of 13.97 during JJ2001-2011 compared to 17.6 percent at the all India level. In fact, the state has achieved replacement level of fertility; the TFR in Odisha was 2.0 in 2016 as per the Sample Registration System (SRS). The latest round of the National Family Health Survey (NFHS-4, 2015-16) recorded a TFR of 2.1 in Odisha. On the other hand, mortality during early childhood period is substantially high. Even so far as female literacy rate goes the national average rate is higher than that of Odisha. The infant mortality rate as well as antenatal check up in Odisha had an edge over India as well. Even so far as female literacy rate goes the national average rate is higher than that of Odisha.

Table 1: Socio-economic, Demographic and Health Indicators in Odisha and India

Indicators	Odisha	India
Population (million)*	41.9	1210.2
Share of Population (%) *	3	—
Percent of Population in ages 0-6 population	12	13.1
Child sex ratio(female per thousand male in 0-6 age group)*	934	914
Sex ratio*	978	940
Percentage of People Living Below Poverty Line (BPL)	32.	21.9
Level Urbanisation (%) *	17	27.8
Area (sq. km)*	155,707	3,287,240
Population density (ersons per square km)*	269	382
Scheduled Caste population(million) *	7.2	201.3
Scheduled Tribe population (million) *	9.6	104.2
Proportion of SC population (%)*	16.5	8.2
Proportion of ST population (%) *	22.1	16.2
Literacy rate (%) *	73.45	74.4

Female literacy rate (in %) *	64.4	74.0
Total fertility rate**	2.05	2.18
Crude Birth Rate (CBR) ****	20.5	20.4
Crude Death Rate (CDR) ****	7.8	6.4
Infant Mortality Rate**	39.6	40.7
Under Five Mortality Rate**	48.1	130
Maternal Mortality Ratio (MMR) Per 100,000 live births+	180	130
Underweight (children of age below five years)**	34.4	42.5
Stunting (children of age below five years)**	34.1	48.0
Wasting (children of age below five years)**	20.4	19.8
% of women with full Antenatal Coverage**	94.0	84.0

Sources: * Census, 2011: India, Registrar General (2011), ** NFHS-4: International Institute of Population Sciences and ICF (2017), *** NSSO, 2011-12: Government of India (2013), **** SRS, 2016: India, Registrar General (2016) and + SRS, 2014-16: India, Registrar General (2018)

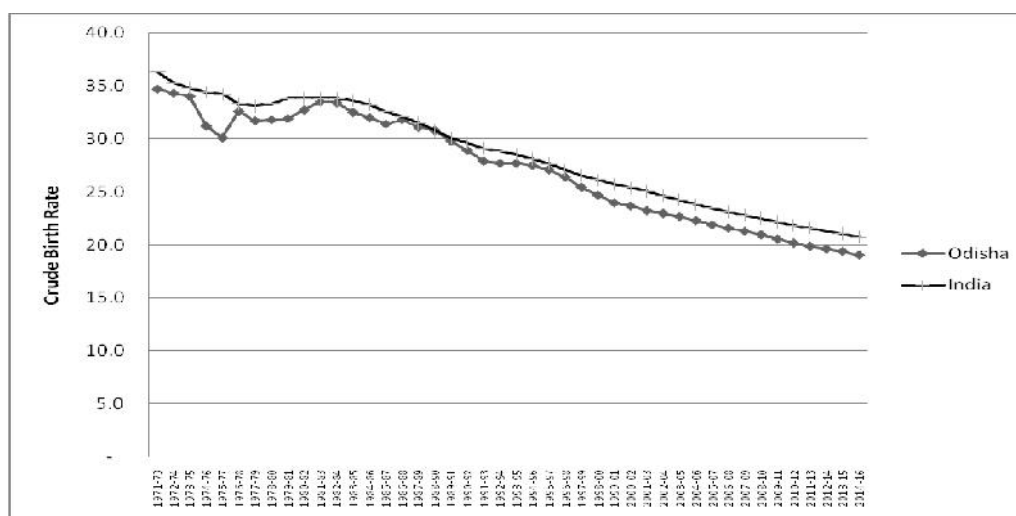
Fertility decline over time

The CBR declined significantly from 34.6 in 1971 to 18.6 in 2016 in Odisha. Similarly, the TFR declined from 4.7 in 1971 to 2.0 in 2016. Compared to the Odisha's figure, national average for CBR declined from 36.9 in 1971 to 20.4 in 2016, and the TFR declined from 5.2 in 1971 to 2.3 in 2016. If we compare the trend line of both CBR and TFR for India as whole with

Odisha, it can be observed that the pace of decline in fertility rate in Odisha has been slightly slower than the average decline in the country. However, the fertility rate has been lower in Odisha than in India from 1990 onward (fig 1 and 2).

If one looks at the data on the CBR and TFR in Odisha over the last five and half a de-

Fig 1: Crude Birth Rate in Odisha as Compared to National Figure, 1971-2016 (Based on Three Year Moving Average)

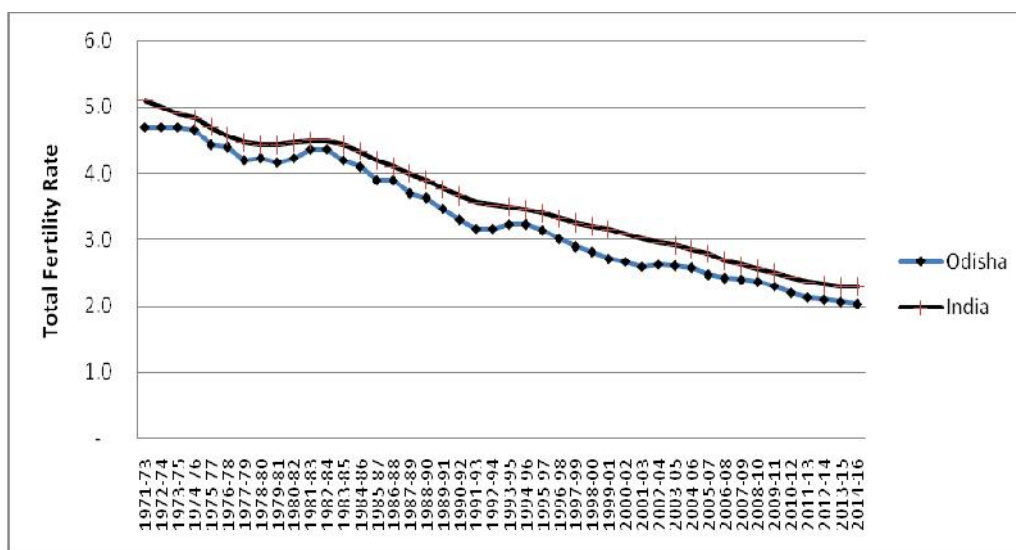


Source: Based on SRS data

decades one can find some significant ups and downs, particularly in the 1970's, 80's and 90's. Even when one takes three-year moving averages to smooth annual fluctuations, one finds similar fluctuations in the 1970's and 80's. From the three years moving average of CBR and TFR for the state of Odisha and the country as a whole, it is observed that national figures for rural and urban are higher than that of Odisha.

Though the state of Odisha has achieved the replacement level of fertility (TFR 2.1) in 2013, the country as whole has not achieved yet, indicating the heterogeneities in fertility level across states. Note that as per the SRS data, TFR in Odisha was 2.0 while the national figure stood at 2.3 (See appendix 1 and 2 for both annual and three-year moving average data).

Fig 2: Total Fertility Rate in Odisha as Compared to National Figure, 1971-2016 (Based on Three Year Moving Average)



Source: Based on SRS data

Table 2: Trend in Fertility Rate in Odisha for the five-year period preceding the Survey: Evidences from NFHS Surveys

NFHS Rounds	CBR			TFR		
	Total	Rural	Urban	Total	Rural	Urban
NFHS-1 (1992-93)	26.50	27.00	23.90	2.92	3.00	2.53
NFHS-2 (1998-99)	24.30	22.40	20.10	2.46	2.50	2.19
NFHS-3 (2005-06)	22.10	23.00	17.70	2.37	2.48	1.89
NFHS 4 (2015-16)	18.10	18.70	15.60	2.05	2.12	1.73

Source: IIPS and ICF, 'National Family Health Surveys 4, Orissa'. 2015-16

All the four rounds of the National Family Health Survey (NFHS 1 to 4) also reveal a de-

clining trend of CBR and TFR in Odisha. During the period between first and second survey

(1992-93 and 1998-99), CBR declined by 2.2 points and TFR declined by 0.5 points. CBR and TFR reduced by 2.2 points and 1 point in Odisha during the second inter-survey period (during 1998-99 to 2005-06). Further, the period between 2005-06 and 2015-16 witnessed decline of both CBR and TFR by 3 points and 0.3 points respectively. It is clear that Odisha was close to the replacement level of fertility in the first part of the first decade of the current century. As SRS data reveal the state achieved the replacement level in 2013. Further, the TFR figure for Odisha has fallen below replacement level in 2015-16.

Table 3: Total Fertility Rate by Background Characteristics

Background Characteristics	Total Fertility Rate	Total Wanted Fertility Rate
Residence		
Urban	1.73	1.50
Rural	2.12	1.72
Schooling of the Mother		
No Schooling	2.66	2.07
< 5 Years Complete	2.42	1.97
5-9 years complete	2.03	1.74
10-11 years complete	1.85	1.62
12 years or more complete	1.60	1.48
Religion		
Hindu	2.04	1.68
Muslim	2.00	1.61
Christian	2.38	1.80
Caste/Tribe		
Scheduled Caste	2.13	1.62
Scheduled Tribe	2.46	1.91
Other Backward Class	1.87	1.62
Other	1.81	1.58
Don't Know	2.16	1.81
Total	2.05	1.69

Note: Total Fertility Rate for the three years preceding the survey (NFHS 4, 2015-16)

Source: IIPS and ICF, 'National Family Health Surveys 4, Orissa'. 2015-16

The TFR in urban areas is well below replacement level and fertility is at replacement level even in rural areas (2.1 children per women). Differentials in fertility by schooling of mother and caste/tribe are wide. The gradient is clear too; higher the years of schooling lower the fertility. Women with no schooling have 2.7 children while the figure for the women with 12 or more years of schooling is 1.1. Among the social groups, the TFR figure is highest among

Scheduled Tribe (ST) communities, followed by Scheduled Caste (SC), Other Backward Classes and Other castes.

Table 3 also indicates that Total Wanted Fertility Rate (TWFR) is lower as compared to the TFR in Odisha irrespective of socioeconomic backgrounds. The TWFR in Odisha is 1.69 which is well below 2.1. Obstacles to use contraceptives and social pressures to have more children are generally cited as the reasons for

the gap between TFR and TWFR. Percentage of currently married women age 15-49 with unmet need of family planning in Odisha is 13.6 % (IIPS and ICF, 2017). The figure substantiates the hypothesis concerning the influence of obstacles to contraceptive use on the lower TWFR than TFR. Of course, no one can deny

the social pressures from the elders in the family on reproductive choice of the couples. Generational change coupled with increase contraceptive (particularly modern) prevalence rate would certainly bring TFR well below the replacement level.

Trends in mortality and life expectancy

Crude death rate (CDR)

In 1970s the Crude Death Rate (CDR) figures in Odisha had fluctuated around 14 to 20 with the highest figure (20) recorded in 1972. There has been a secular trend in the decline of CDR since mid 1980s. Towards the end of the

twentieth century, the CDR figure was close to 10 and it has reached 7.8 in 2016. The national figure for CDR stood at 6.4 in 2016. During the period between 1971 and 2016, the national figure continues to be lower than that of Odisha (figure 3).

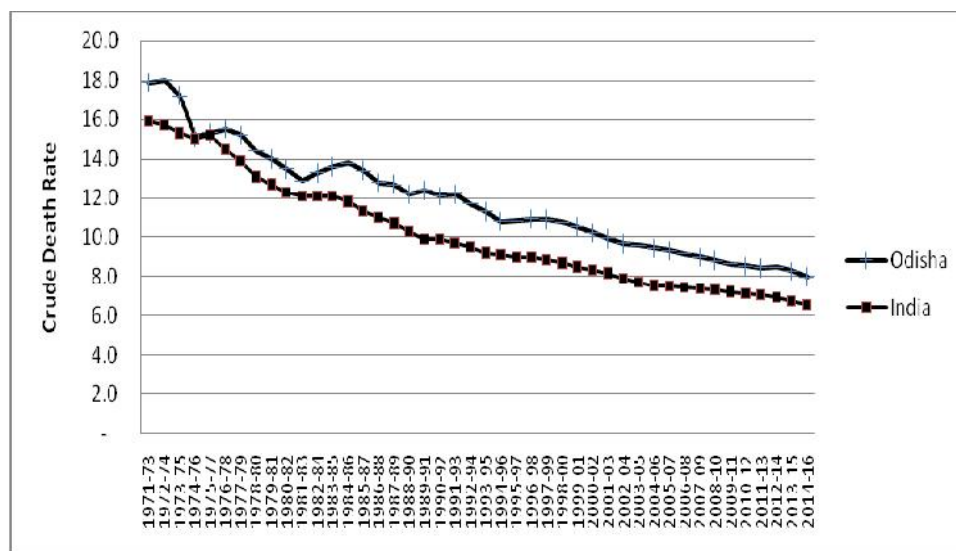


Fig 3: Crude Death Rate in Odisha as Compared to National Figure, 1971-2016 (Based on Three Year Moving Average)

Source: Based on SRS data

Infant mortality rate

Infant Mortality Rate (IMR) data of Odisha prior to independence are scarce. However, data from annual reports of public health commissioner with the Government of India for the period 1940-44, do give an idea about over all Infant mortality situations in the state (Table 4). If

the source is to be believed, average IMR of rural Odisha during 1937-41 was 209.3 compared to 172.5 in urban Odisha. The table 4 shows that IMR in 1943 and 1944 rose significantly from the figure of 1942. This rise in infant death can be attributed to the occurrence

of famine during 1943-44. Figures from 1944 to 1946 show a gradual decline in both for rural and urban Odisha. In 1946, the recorded IMR figures for rural and urban Odisha were 172.6 and 144.9 respectively. It can be noticed that during the period 1937-46, IMR in urban Odisha

was comparatively lower than that of rural Odisha. While comparing these 1946 figures with 1971 estimates of SRS (129), it can be observed that there had been decline in IMR during this time span.

Table 4: Infant mortality rate (IMR) per 1000 live births in Orissa during 1937-46

Place of Residence	Average	1937-41	1942	1943	1944	1945	1946	1971*
Rural	209.3	191.3	203.1	209.4	178.4	172.2	131	
Urban	172.5	167.2	187.6	172.6	168.2	144.9	84	

Source: Government of India (1947), Statistical Appendices to Annual Reports of Public Health Commissioner with the Government of India for the Period 1940-44, Delhi: Government Press (cited in Maharatna, Arup.1996. The Demography of Famines, Oxford University Press, Delhi, page no. 226)

* Sample Registration System (SRS)

Infant mortality rate has declined significantly in India from 129 in 1971 to 34 in the year 2016. Compared to this national average, Odisha's IMR declined from 127 in 1971 to 44 in 2016. In 1971, infant mortality rate in Odisha was nearly same as in India (129 for India and 127 for Odisha), but surprisingly infant mortal-

ity rate in Odisha is much higher than that of India in 2016. So it is quite visible that the pace of decline is much lower in Odisha compared to the national average. This trend line for Odisha is compared with the trend for national average for the same period. Figure 4 shows the decline in Odisha's IMR for the period from 1981 to

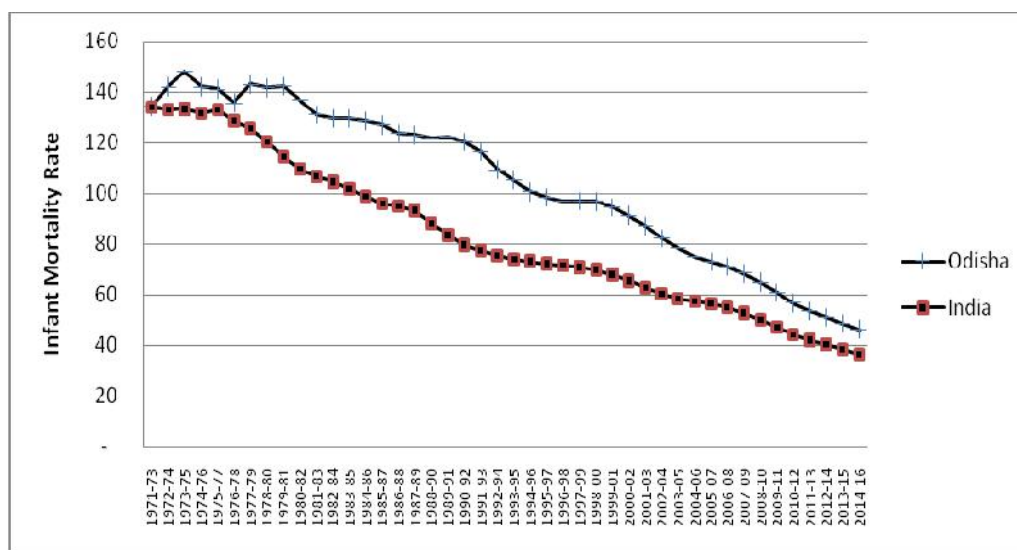


Fig 4: Infant Mortality Rate in Odisha as Compared to National Figure, 1971-2016 (Based on Three Year Moving Average)

Source: Based on SRS data

2016, the last year for which SRS data have been published. It indicates that the decline in infant mortality in Odisha has been slightly slower than the average decline in the country. If one looks at the data on the IMR in Odisha over the last four decades, one deciphers some significant up-and-down movements particularly in the 1970's and 80's. Even when one takes three-year moving averages to smooth annual fluctuations, one finds similar fluctuations only in 70's. The rate was 134 in 1972, climbed to its

peak at 148 in 1974, then there was a fall in this rate until around 1977, followed by a small rise and during 78-80 the figure fluctuated between 143 and 142. Since then it has started falling till 2016. The general pattern of movement in the IMR in India as a whole is quite different, as decline in IMR has been smoother during the last four and half a decades. In contrast, annual data for Odisha shows fluctuations in 1970's (see appendix for annual and three year moving average data).

Table 5: Annual Rate of Change in Infant Mortality Rate in India and Orissa

Country/state	1971-81	1981-91	1991-2001	2001-2011	2011-2016
Odisha	0.62	-0.81	-2.66	-3.73	-4.56
India	-1.47	-2.72	-1.75	-3.33	-4.54

Source: Calculated from SRS Annual Estimates

The rates of decline in IMR in Odisha in 1970s and 1980s were substantially lower than that of national average. In fact, IMR increased during 1971-81 (annual rate of change was 0.62). A steady but slow decline occurred during the 1980s in Odisha and even faster rate of decline has been noticed in 1990's and in the first decade of the twenty first century com-

pared to the previous two decades (Table 5). The rate of annual rate decline after 1990 in Odisha was also quite higher than the figure for India. One could argue that the present higher level of Infant Mortality Rate in Odisha compared to the national average can be attributed to the slower decline in 1970s and 1980s (Das, 2011).

Table 6: Trend in Infant Mortality Rate in Orissa and India for the five-year Period preceding the Survey: Evidences from NFHS Surveys

Survey Year	IMR Orissa	UFMR India	Orissa	India
NFHS-1(1992-93)	112	79	131.0	109.3
NFHS-2(1998-99)	81	68	104.4	94.9
NFHS-3(2005-06)	65	57	90.6	74.3
NFHS-4(2015-16)	40.1	28.5	48.1	34.4

Source: IIPS and ICF, 'National Family Health Surveys 4, Orissa'. 2015-16

As per NFHS-4 estimates IMR was 40 in Odisha which is higher than the national figure (28.5) in 2015-16. During NFHS 1 and NFHS 4, the rate of decline in IMR in Odisha has been higher as compared to the whole country (table-6).

Under five Mortality

All the rounds of the National Family Health Survey also provide an overview of trend of U5MR in Odisha over the last two and half a decades (Table 6). The under-five mortality rate has declined from 136.9 deaths per 1000 live

births during 1992-93 to 48.1 in 2015-16. Over the last one decade (during NFHS-3 to NFHS-4), the UFMR has declined to the tune of around 42 points; it declined by around 40 points during NFHS 1 (1992-93) and NFHS 3 (2015-16). Though UFMR has declined at accelerated rate in Odisha, it remains substantially high, higher than the national figure.

Maternal mortality

Maternal Mortality Ratio (MMR) during 2011-13 was 222 in Odisha compared to the national figure of 167 (Table 7). Among EAG states, there is a great deal of diversity in the levels and pace of decline. According to SSR estimates for 2011-13, the MMR figure for

Odisha is lower than Uttar Pradesh/Uttarakhand (285) and Rajasthan (244) while Bihar/Jharkhand (208) and Madhya Pradesh/Chhattisgarh (221) recorded a lower MMR as compared to Odisha. Though all EAG states have experienced a decline in MMR, the pace of decline in MMR has been slower in Odisha as compared to other EAG states. Note that during 1997-98, MMR in Odisha was lower than the figures for rest EAG states. Barring Assam (300), the MMR in all Non-EAG states is much below the figure of Odisha. For example, the MMR figures for Kerala and Maharashtra are 61 and 68 respectively.

Table 7: Trend in Maternal Mortality Ratio in Orissa as compared to other states

India & Bigger States	Period						
	1997-98	1999-01	2001-03	2004-06	2007-09	2010-12	2011-13
INDIA TOTAL	398	327	301	254	212	178	167
Assam	568	398	490	480	390	328	300
Bihar/Jharkhand	531	400	371	312	261	219	208
Madhya Pradesh/Chhattisgarh	441	407	379	335	269	230	221
Odisha	346	424	358	303	258	235	222
Rajasthan	508	501	445	388	318	255	244
Uttar Pradesh/Uttarakhand	606	539	517	440	359	292	285
EAG AND ASSAM SUBTOTAL	520	461	438	375	308	257	246
Andhra Pradesh	197	220	195	154	134	110	92
Karnataka	245	266	228	213	178	144	133
Kerala	150	149	110	95	81	66	61
Tamil Nadu	131	167	134	111	97	90	79
SOUTH SUBTOTAL	187	206	173	149	127	105	93
Gujarat	46	202	172	160	148	122	112
Haryana	136	176	162	186	153	146	127
Maharashtra	166	169	149	130	104	87	68
Punjab	280	177	178	192	172	155	141
West Bengal	303	218	194	141	145	117	113
Other		276	235	206	160	136	126
OTHER SUBTOTAL	184	229	199	174	149	127	115

Source: India, Registrar General (2013) SRS Compendium.

Life Expectancy

Table 8 and figure 5 shows the trend in life expectancy at birth in Odisha. Life expectancy at birth in Odisha during the period 2010-13 was 65 lower than national average (68) and many states of the country. Apart from Madhya Pradesh (including Chhattisgarh) and Uttar Pradesh, all other EAG states have higher life expectancy than Odisha. However, the life expectancy in Odisha has increased from as low as 46 years during 1970-75 to 65 years during 2010-13, registering a rise of 19 years in the expectancy during the last four and a half decades. Life expectancy has increased both in rural and urban Odisha. Rural-Urban gap in life expectancy was 10 years during 1970-75, de-

creased to 3 years during 2010-13. Though Female expectancy was below the male expectancy before 1980, the gap in favour of females has been noticed since early 1980s. The latest data show a life expectancy of 65.8 for males and 69.3 for females during 2010-13. A comparison with neighbouring states tells us that life expectancy in Odisha is higher than Madhya Pradesh (including Chhattisgarh) and lower than the states of West Bengal, Andhra Pradesh and Bihar (including Jharkhand) (Table 9). Among the rest of the major states of India, while the figure for Odisha is lower than that of the relatively developed states of India, Rajasthan, Uttar Pradesh and Assam have lower life expectancy as compared to Odisha.

Table 8: Trends in Life Expectancy at Birth, Odisha

Period	ALL			Rural			Urban		
	Total	Male	Female	Total	Male	Female	Total	Male	Female
1970-75	46	46	45	45	46	45	55	55	56
1976-80	49	50	48	49	49	48	58	58	59
1981-85	53	53	53	52	52	52	60	59	61
1986-90	54	55	54	54	54	53	62	61	63
1991-95	57	57	56	56	56	55	64	62	67
1996-00	58	58	59	58	57	58	64	63	64
2001-05	61	60	62	60	59	61	66	65	68
2006-10	63	62	64	62	62	63	67	66	69
2009-13	65	64	66	64	63	65	69	68	70

Source: India, Registrar General (2013) SRS Compendium.

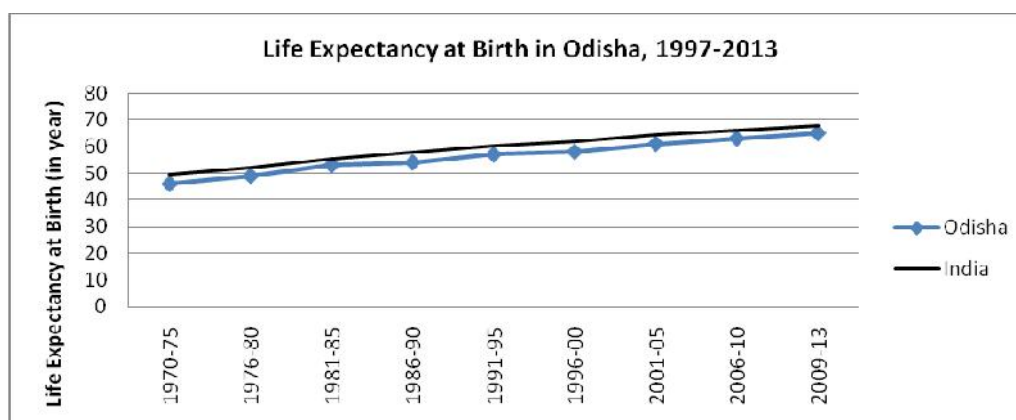


Fig 5: Life Expectancy in Odisha as Compared to National Figure, 1997-2013

Table 9: Trends in Life Expectancy at Birth in Odisha ass compared other states

Year	1970-75	1976-80	1981-85	1986-90	1991-95	1996-00	2001-05	2006-10	2009-13
Odisha	46	49	53	54	57	58	61	63	65
Andhra Pradesh	48.8	53.1	58.4	59.1	61.8	62.7	65	65.8	67.9
Bihar (including Jharkhand)								65.8	67.7
Madhya Pradesh (Including Chhattisgarh)	47.2	49	51.6	53	54.7	57.1	59.7	62.4	63.8
West Bengal			57.4	60.8	62.1	64.3	67.2	69	69.9
Rajasthan	48.4	51.9	53.5	55.2	59.1	62.1	64.5	66.5	67.5
Uttar Pradesh	43	46.2	50	53.4	56.8	59.2	60.8	62.7	63.8
Assam	45.5	51.1	51.9	53.6	55.7	57.4	59.2	61.9	63.3
Maharashtra	53.8	56.3	60.7	62.6	64.8	65.9	68	69.9	71.3
Punjab	57.9	60.5	63.1	65.2	67.2	66.5	68.8	69.3	71.1
All India	49.7	52.3	55.5	57.7	60.3	61.9	64.3	66.1	67.5

Source: India, Registrar General (2013), SRS Compendium 2013

Population growth and changes in age structure

In India, the first regular Census of population started in 1881, which recorded a population of 6.1 lakh and increased to 5.7 in 1991. The first census of the 20th century (the 1901 census) observed a population size of 10.3 lakhs in Odisha. Sinha (1958) by analysing the population growth in Odisha during 1871-1951 observed that 'the unprecedented rise in absolute figures and phenomenally at a faster rate towards the later few decades is striking. The total population has not only become a little less than threefold but has recorded 186.2 percent rise in 1951 over the figures of 1881. The growth when analysed in decades, reveals, loss in total population during 1911-21 in Orissa. Trend in population when viewed against a background of all India figures presents a different picture. The absolute population in India does not indicate a threefold increase but has only recorded 66.1 percent growth in 1951 over the

figures of 1881. The trend, when, analysed in decades indicates, that the decades of marked increase have regularly alternated with decades of slight increase and the growth is sporadic. But from 1921 the growth in successive decades is rapid'. The present study uses 1901 as the base year to analyse the growth of population in Odisha. The absolute size of population along with decadal growth rate of population since 1901 is presented in Table 10. As evident from the table 10, negative growth rate of population was recorded during 1911-21. "Besides the global war, the devastating floods, famines and epidemics in different parts of the State brought down the number of people in Orissa as a chain event compared with the previous decadal counting"(Mahapatra, 2004: 35). Epidemics like small pox, cholera and dysentery during this decade were common in Odisha. More importantly, starting from 1917, influenza had spread its tentacles up to the year 1919. It took away a

lot of precious lives putting the rates of morality at a higher level. Besides that ravaging flood of 1917, the drought of 1918 worsened the living conditions of the people to a large extent. Coupled with bad harvest for some years, these vagaries of nature also demanded their share

and created a condition of scarcity of foodstuff resulting in high market prices which led to starvation deaths. In absolute terms the 11,378,875 people in Orissa in 1911 was reduced to 11,158,586 in 1921 downsizing by about two lacs and 20 thousand people in 10 years time.

Table 10: Population Growth in Odisha (1901-2011)

Census	Population	Change in Population Size	Decadal Variation (in Percentage)
1901	10,302,917		
1911	11,378,875	1,075,958	10.44
1921	11,158,586	- 220,289	- 1.94
1931	12,491,056	1332,470	11.94
1941	13,767,988	1,276,932	10.22
1951	14,645,946	877,958	6.38
1961	17,548,846	2,902,900	19.82
1971	21,944,615	4,395,769	25.05
1981	26,370,271	4,425,656	20.17
1991	31,569,736	5,289,465	20.06
2001	36,804,660	5,144,924	16.25
2011	41,947,358	5,142,698	13.97

Source: India, Registrar General of India (2011). Census of India 2011

The highest decadal growth rate (25.05 %) was recorded during the decade 1961 - 1971. Since then the growth rate has been declining. During the current decade 2001 - 2011, Odisha has registered a growth rate of 13.97 percent compared to the growth rate of 16.25 percent the decade 1991 - 2001. Fig 6 shows the steady decline in natural growth (CBR-CDR) in 1970s and 1980 and the somehow faster rate of decline since 1990s. This has happened due to the higher rate of decline in the CBR as compared to the CDR. During the same period, TFR declined at a higher rate as compared to the earlier period too. Certainly, the population growth pattern clearly corresponds to the rate of decline in natural growth. At the same time, migration as a component of population change has not played a major role in increase in population in Odisha as out-migrants from the state

out- numbered in-migrants in the past. Odisha (Das, 2012).

Declining fertility along with improvement in life expectancy in Odisha has changed the population age-structure. Fig 7 and 8 are clearly showing the shrinking base of the pyramid with bulge moving upwards. The broad age group 15-59 generally labelled as working ages has gained by over three points in its share during 2001 to 2011, from 58.4 percent to 61.5 percent. The percentage of population in the young age group (0-14) declined from 33.2 to 28.8 percent, registering a decline by 4.4 points (Table 11). The old ages (60 and above) have also gained but only marginally, from 8.3 to 9.5 percent. Ageing has begun to occur in Odisha. The most conspicuous changes are seen in the shares of the young age groups.

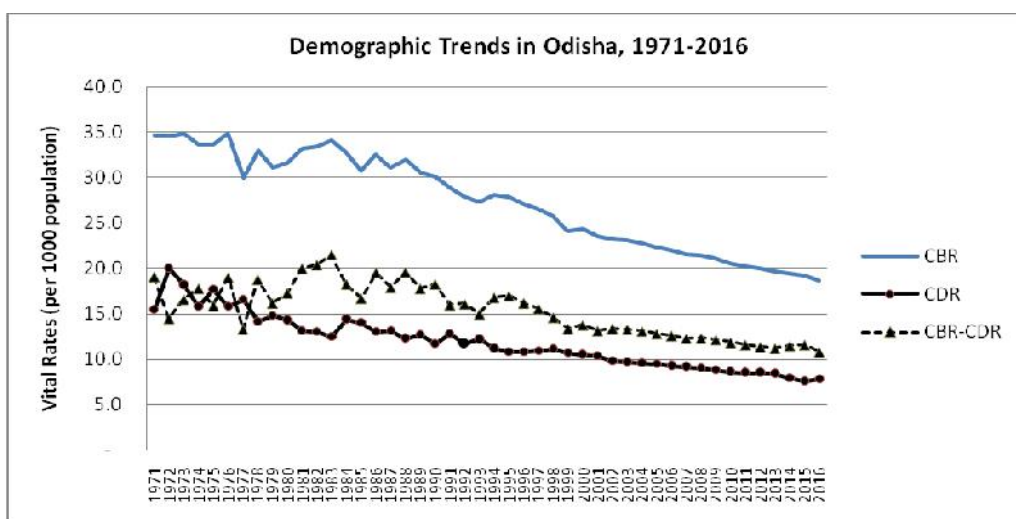


Fig 6: Demographic Trends in Odisha, 1971-2016

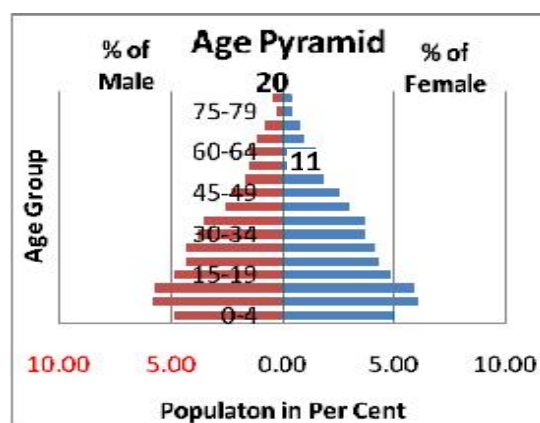
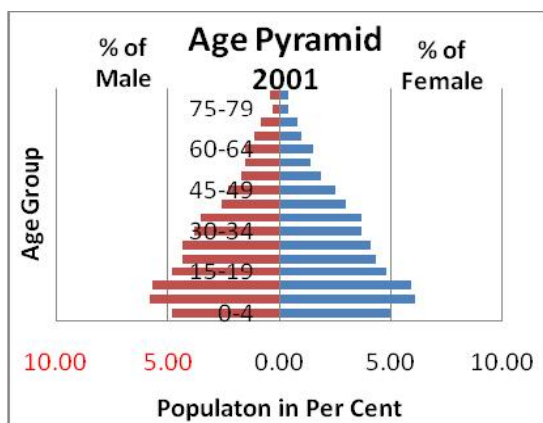


Fig 7: Population Pyramid of Odisha, 2011 Fig 8: Population Pyramid of Odisha, 2011

Table 11: Percentage Distribution of Population in Different Age Groups in Odisha, 2011

Age group	Odisha		India
	2001	2011	2011
0-14	33.2	28.8	30.76
15-59	58.4	61.5	60.29
60+	8.3	9.5	8.58

Source: Calculated Based on Census 2011 C.D.ROM

Table 12: Trends in Dependency Ratio in Odisha, 2001-2011

Dependency Ratio	Odisha		India
	2001	2011	2011
Child Dependency Ratio	56.8	46.8	51.01
Old Dependency Ratio	14.1	15.5	14.22
Total Dependency Ratio	70.9	62.3	65.23

Source: Calculated Based on Census 2011 C.D.ROM

A positive consequence of the change in the age distribution is that the dependency ratio has fallen, from 70.9 percent to 62.3 percent (Table 12). This is accounted for by the drop in the young age dependency ratio from 56.8 percent to 4.8 percent. On the other hand, old age dependency ratio increased from 14.1 to 15.5 percent. The decreasing share of children age group is certainly a result of recent fertility decline in the state. Clearly, the burden of supporting children has come down as couples now have fewer children than in the past.

Regional variations in transition

The CBR in the four districts viz., Rayagada, Koraput, Nabarangpur and Malkangiri is more than 25 while Jagatsingpur, Puri, Cuttack, Khorda, and Kendrapara have recorded relatively lower CBR. The broad regional pattern shows that the level of CBR is high in the south-

ern parts of Odisha and low level of CBR is recorded in the coastal part and industrially prosperous north-western districts of Odisha. Like CBR, low level of TFR is seen in coastal districts of Odisha where as high rate is seen mainly in the southern districts. Infact, TFR is well below replacement level in four districts, Jagatshingpur, Cuttack, Khorda and Puri. Kendrapara has TFR below replacement too (2.0). Bargarh, Jharsuguda and Sambalpur TFR is 2.1. In two districts (Nabarangpur and Malkangiri) of the state TFR increased between 2001 and 2011. Coefficient of Variation (CV) values indicate a slight increase in the degree of regional variation in fertility during 2001 and 2011. It is to be noted that the CV value for CBR increased from 13.59 in 2001 to 19.45. Similarly CV value for TFR increased from 12.67 in 2001 to 19.10 in 2011

State & Districts	CBR		Decadal change in percentage in CBR	TFR		Decadal change in percentage in TFR	Decadal Population Growth Rate, 2001-2011
	2001	2011		2001	2011		
Odisha	23.6	19.6	-16.9	2.8	2.4	-14.2	13.97
Bargarh	20.6	16.8	-18.4	2.5	2.1	-16	9.9
Jharsuguda	21.1	16.9	-19.9	2.6	2.1	-19.2	12.
Sambalpur	21.2	17.7	-16.5	2.6	2.1	-19.2	12.2
Debagarh	25.5	20.5	-19.6	3.1	2.5	-19.3	13.9
Sundergarh	22.8	19.3	-15.3	2.7	2.3	-14.8	13.7
Kendujhar	25.3	23.1	-8.6	3.0	2.8	-6.6	15.4
Mayurbhanj	26.0	21.4	-17.6	3.0	2.7	-10	13.1
Baleshwar	25.2	19.1	-24.2	2.9	2.3	-20.6	14.5

Bhadrak	24.8	18.9	-23.7	2.9	2.3	-20.6	13.0
Kendrapara	21.8	17.0	-22.0	2.6	2.0	-23.0	10.6
Jagatsingpur	18.8	14.3	-23.9	2.3	1.7	-26.0	7.4
Cuttack	19.6	15.4	-21.4	2.4	1.8	-25	11.9
Jajpur	21.8	18.1	-16.9	2.6	2.2	-15.3	12.4
Dhenkanal	21.8	18.1	-16.9	2.7	2.1	-22.2	11.8
Anugul	23.4	18.5	-20.9	2.9	2.2	-24.1	11.6
Nayagarh	20.9	17.2	-17.7	2.5	2.1	-16	11.3
Khurda	20.3	16.2	-20.1	2.4	1.9	-20.8	19.6
Puri	20.2	15.8	-21.7	2.4	1.8	-25	13.0
Ganjam	24.0	18.4	-23.3	2.9	2.2	-24.1	11.4
Gajapati	27.6	24.1	-12.6	3.3	2.9	-12.1	11.0
Kandhamal	30.8	24.7	-19.8	3.6	3.0	-16.6	12.9
Boudh	27.4	22.6	-17.5	3.2	2.8	-12.5	17.8
Subarnapur	22.7	19.2	-15.4	2.8	2.5	-10.7	20.4
Balangir	22.9	21.2	-7.4	2.8	2.7	-3.5	23.3
Nuapada	25.9	23.1	-10.8	3.0	2.9	-3.3	14.3
Kalahandi	26.8	23.0	-14.1	3.2	2.8	-12.5	17.8
Rayagada	28.5	25.1	-11.9	3.3	2.9	-12.1	15.7
Nabarangpur	30.0	28.6	-4.6	3.4	3.5	2.9	18.8
Koraput	27.3	26.2	-4.0	3.1	3.0	-3.2	16.6
Malkangiri	28.8	30.0	4.1	3.3	3.5	6.0	21.5

Source: Guilmoto and Rajan (2002). District Level Estimates of Fertility from India's 2001 Census.

Guilmoto and Rajan (2013). Fertility at District Level in India: Lessons from the 2011 census.

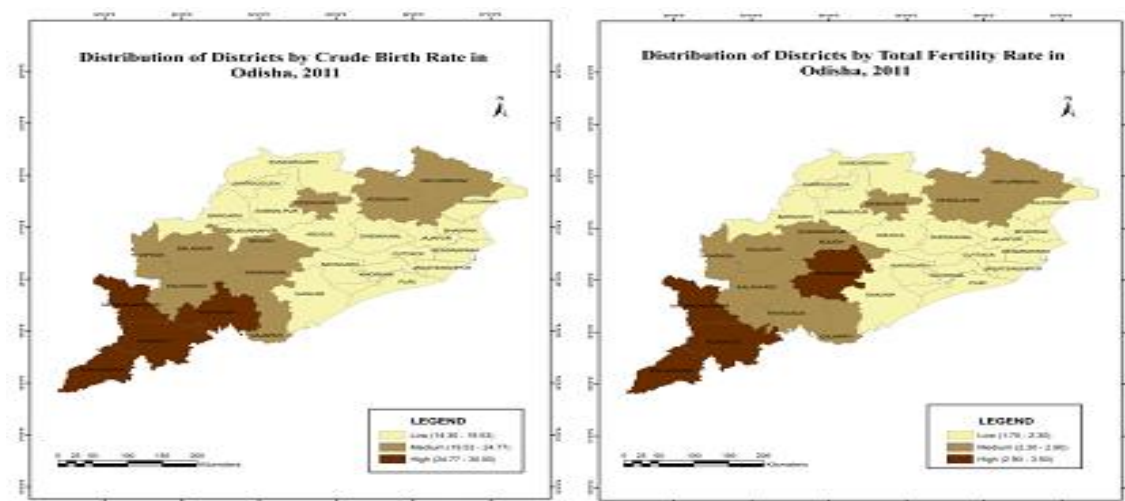


Fig 9: Crude Birth Rate in Odisha, 2011 Fig 10: Total Fertility in Odisha, 2011

Higher rate of decline in CBR is observed in the coastal districts of the state such as Balasore, Jagatsingpur, Bhadrak, Ganjam, Kendrapara, Puri, Cuttack, Angul and Khordha during 2001-2011 while it declined moderately in Kandhamal, Deogarh, Bargarh, Nayagarh. Southern and south-western districts, such as Koraput, Nabarangpur, Balngir, Kendujhar, Nuapada, Rayagada, Gajapati and Kalahandi experienced a relatively lower rate of decline. Malkangiri is the only district where CBR has increased between 2001 and 2011. Similar pattern of change is also observed in case of TFR. Regional patterns in population growth rate correspond to regional pattern of fertility decline.

Conclusions

Odisha's performance in the demographic front amidst poor socio-economic scenario has been impressive. Fertility transition in Odisha has been underway for the past four decades. This transition has covered all regions and sections of the society during the last decade, though to varying degrees. All agricultural prosperous coastal and industrially developed north-western districts have low fertility. Infact, five coastal districts along with three non-coastal districts have already reached low replacement level fertility while the tribal populated southern districts and few northern districts are yet to break the barrier. Though mortality indicators have shown declining trends, the current levels are substantially high. Southern region is lagging behind the coastal and northern regions of Odisha. However, a scenario of low birth rate along with poor economic conditions and high childhood mortality remains a matter of concern. Consequent on the decline in birth rate and death rate, Odisha has experienced continuous decline in population growth in Odisha.

The decreasing share of children population is certainly a result of recent fertility decline in the state. In broad terms, the age distribution has shifted towards middle ages, with fall

in the share of the child population and a corresponding rise in the share of adults. Clearly, the burden of supporting children has come down. Ageing has begun to occur in Odisha with the marginal increase in share of old age people in the population. With the state approaching towards a low level fertility and mortality situation, Odisha would definitely be a high performing state in demographic indicators within the context of relatively poor performing social and economic variables. At the same time, the current age structure of Odisha's population with substantially high proportion of population of working age group provides windows of opportunities for economic development. To harness the demographic dividend, the state as well as union government needs to invest generously on health and education.

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Appendix 1: Demographic Indicators in Odisha as Compared to National Figure, 1971-2016 (Annual figures)

	CBR		CDR		Natural Change		IMR		TFR	
	Odisha	India	Odisha	India	Odisha	India	Odisha	India	Odisha	India
1971	34.6	36.9	15.5	14.9	19.1	22.0	127	129	4.7	5.2
1972	34.5	36.6	20.0	16.9	14.5	19.7	131	139	4.6	5.2
1973	34.8	34.6	18.2	15.5	16.6	19.1	145	134	4.8	4.9
1974	33.6	34.5	15.8	14.5	17.8	20.0	150	126	4.7	4.9
1975	33.6	35.2	17.7	15.9	15.9	19.3	149	140	4.6	4.9
1976	34.8	34.4	15.8	15.0	19.0	19.4	127	129	4.7	4.7
1977	29.9	33.0	16.6	14.7	13.3	18.3	147	130	4.0	4.5
1978	32.9	33.3	14.1	14.2	18.8	19.1	133	127	4.5	4.5
1979	31.0	33.1	14.8	12.8	16.2	20.3	149	120	4.1	4.4
1980	31.6	33.3	14.3	12.4	17.3	20.9	143	114	4.1	4.4
1981	33.1	33.9	13.1	12.5	20.0	21.4	135	110	4.3	4.5
1982	33.4	33.8	13.0	11.9	20.4	21.9	132	105	4.3	4.5
1983	34.0	33.7	12.5	11.9	21.5	21.8	126	105	4.5	4.5
1984	32.7	33.9	14.4	12.6	18.3	21.3	131	104	4.3	4.5
1985	30.7	32.9	14.0	11.8	16.7	21.1	132	97	3.8	4.3
1986	32.5	32.6	13.0	11.1	19.5	21.5	123	96	4.2	4.2
1987	31.0	32.2	13.1	10.9	17.9	21.3	126	95	3.7	4.1
1988	31.9	31.5	12.3	11.0	19.6	20.5	122	94	3.8	4.0
1989	30.5	30.6	12.7	10.3	17.8	20.3	121	91	3.6	3.9
1990	30.0	30.2	11.7	9.7	18.3	20.5	122	80	3.5	3.8
1991	28.8	29.5	12.8	9.8	16.0	19.7	124	80	3.3	3.6
1992*	27.8	29.2	11.7	10.1	16.1	19.1	115	79	3.1	3.6
1993*	27.2	28.7	12.2	9.3	15.0	19.4	110	74	3.1	3.5
1994*	28.0	28.7	11.2	9.3	16.8	19.4	103	74	3.3	3.5
1995*	27.8	28.3	10.8	9.0	17.0	19.3	103	74	3.3	3.5
1996*	27.0	27.5	10.8	9.0	16.2	18.5	96	72	3.1	3.4
1997*	26.5	27.2	10.9	8.9	15.6	18.3	96	71	3.0	3.3
1998	25.7	26.5	11.1	9.0	14.6	17.5	98	72	2.9	3.2
1999	24.1	26.0	10.7	8.6	13.4	17.4	97	70	2.7	3.2
2000	24.3	25.8	10.5	8.5	13.8	17.3	95	68	2.8	3.2
2001	23.5	25.4	10.4	8.4	13.1	17.0	91	66	2.6	3.1
2002	23.2	25.0	9.8	8.1	13.4	16.9	87	63	2.6	3.0
2003	23.0	24.8	9.7	8.0	13.3	16.8	83	60	2.6	3.0
2004	22.7	24.1	9.6	7.5	13.1	16.6	77	58	2.7	2.9

2005	22.3	23.8	9.5	7.6	12.8	16.2	75	58	2.6	2.9
2006	21.9	23.5	9.3	7.5	12.6	16.0	73	57	2.5	2.8
2007	21.5	23.1	9.2	7.4	12.4	15.7	71	55	2.4	2.7
2008	21.4	22.8	9.0	7.4	12.3	15.4	69	53	2.4	2.6
2009	21.0	22.5	8.8	7.3	12.2	15.2	65	50	2.4	2.6
2010	20.5	22.1	8.6	7.2	11.9	14.9	61	47	2.3	2.5
2011	20.1	21.8	8.5	7.1	11.6	14.7	57	44	2.2	2.4
2012	19.9	21.6	8.5	7.0	11.4	14.5	53	42	2.1	2.4
2013	19.6	21.4	8.4	7.0	11.3	14.4	51	40	2.1	2.3
2014	19.4	21.0	7.9	6.7	11.5	14.3	49	39	2.1	2.3
2015	19.2	20.8	7.6	6.5	11.6	14.3	46	37	2.0	2.3
2016	18.6	20.4	7.8	6.4	10.8	14.0	44	34	2.0	2.3

* Excludes Jammu & Kashmir due to non-receipt of returns.

Note: 1: Estimates of vital rates at the national level upto 1995 do not include Mizoram as the SRS was not operational in Mizoram till 1995.

2 : Excludes Nagaland (Rural) due to part-receipt of returns from 1995 to 2003.

Source: India, Registrar General, 2013, SRS Compendium, 1971-2013, Government of India, New Delhi.

